

Homopolymer error rate analysis for nanopore amplicon sequencing

Josh Walker

Internal review draft, May 2026 (commit 5d33d1d)

Abstract

We quantify homopolymer (hp) length error rates in Oxford Nanopore amplicon sequencing by base and run length, using two independent sequencing runs of fungal ITS amplicons spanning two basecaller generations (Dorado SUP v3.5 and v5.0) on R10.4.1 chemistry. After filtering biological variants via statistical outlier and bimodal distribution detection, we find that implied per-position error rates at hp lengths 1–5 are consistently below 1.4% for Dorado v5.0 and below 2.3% for v3.5—comparable to the non-hp substitution baseline. Error rates rise steeply at lengths 6+, with G/C 2–4× worse than A/T. We also investigate substitution errors at hp boundaries (interface positions) and find no evidence of excess artifactual variation: well-supported variants at interface positions occur at or below the rate of non-hp variants (rate ratios of 1.03 [95% CI: 0.88–1.19] for v3.5 and 0.88 [0.79–0.99] for v5.0). These results support extending speconsense’s critical error rate (CER) framework to evaluate hp-position variants at lengths 1–5 statistically, rather than discarding them unconditionally via blanket homopolymer normalization. The CER extension should operate at the run level, using empirical error distributions by (base, length) context; the per-position rate serves as a compact parameterization for cross-length comparison, not a mechanistic model of per-base independence.

1 Introduction

1.1 Homopolymer errors in nanopore sequencing

Homopolymer (hp) length errors are the dominant systematic error mode in nanopore sequencing. Approximately 47% of all nanopore sequencing errors originate in homopolymer regions [1], and even with the latest chemistry and basecalling, homopolymer-length indels account for 56% of remaining errors in state-of-the-art genome assemblies [9]. The underlying mechanism is well understood: nanopore sequencing infers base identity from ionic current changes as DNA translocates through the pore, but consecutive identical bases produce no current change, making the length of a homopolymer run difficult to determine from signal alone.

While the qualitative understanding that nanopore sequencing struggles with homopolymers is widespread, quantitative characterization of error rates by homopolymer length and base has been limited. The most comprehensive published data comes from two studies:

- **Sereika et al. (2022)** [6] characterized per-length, per-base hp accuracy at both read and assembly level using a Zymo mock community, with Illumina-polished PacBio HiFi assemblies as ground truth. Their comparison of R9.4.1 and R10.4 chemistry demonstrated a 20–30 percentage point improvement in read-level accuracy at hp lengths 7–9 with the newer dual reader head.
- **Ni et al. (2023)** [5] independently confirmed these trends on human whole genome data, benchmarking R9.4.1 and R10.4 against a T2T reference genome.

Both studies focused primarily on hp lengths ≥ 5 , where errors are dramatic and consequential for genome assembly. Short homopolymers (lengths 1–4) were noted to be highly accurate (>95% at read level) but were not characterized in detail, as they were not considered problematic for typical applications.

The rapid pace of improvement in nanopore basecalling further limits the applicability of published data. Sereika’s R10.4 results used Guppy SUP basecalling; our data spans two basecaller generations (Dorado SUP v3.5.x and v5.0.0) on R10.4.1 chemistry. No published study provides per-length hp error rates for either configuration.

1.2 The speconsense problem

This gap in quantitative understanding has practical consequences for nanopore amplicon analysis. Speconsense’s variant phasing pipeline uses homopolymer normalization to avoid calling false variants caused by hp length errors. Currently, *all* homopolymer length differences are treated as insignificant—a difference between ATC and ATTCC is treated identically to a difference between AAAAAAAAAA and AAAAAAAAAA. This is conservative but lossy: it discards variants that differ only at homopolymer positions, even when those differences are very likely to be biological.

The loss is not merely theoretical. At short hp lengths (1–4), the published literature suggests read-level accuracy exceeds 95%, meaning that a length difference at these positions is far more likely to reflect real biology than sequencing error. Yet speconsense discards this information.

Speconsense’s critical error rate (CER) framework [7] provides a statistical approach for evaluating whether an observed variant is plausibly explained by sequencing error, but it currently applies only to non-hp positions. Extending the CER framework to hp positions requires quantitative error rate estimates by hp length and base—exactly the data that is missing from the published literature for current chemistry. The companion report [8] integrates the rates derived in this analysis into a context-aware CER framework and its implementation in speconsense; the present report provides the empirical foundation.

1.3 This analysis

This analysis quantifies nanopore hp error rates by homopolymer length and base, using real amplicon data from two independent sequencing runs spanning two basecaller generations (Dorado SUP v3.5 and v5.0), to determine whether the CER framework can be extended to evaluate hp-position variants statistically rather than discarding them unconditionally. We also investigate whether substitution errors at homopolymer boundaries (interface positions) require special treatment, or can be evaluated using the same framework as non-hp positions.

2 Methods

2.1 Data

We analyzed specimens from sequencing runs of fungal ITS amplicons (~700 bp), all produced by Mycota Lab (Hoosier Mushroom Society) following the protocol of Russell [4] and sequenced on Oxford Nanopore R10.4.1 E8.2 chemistry. Reads were demultiplexed with specimux and clustered with speconsense using `--presample 0 --max-sample-size 10000` to avoid quality-sorted presampling bias.

- **ont37:** 94 qualifying specimens from 651 with c1 MSAs (environmental specimens). Basecalled with Dorado SUP v3.5.1 (70% of reads) and v3.5.2 260bps (30%), sequenced February

2024. Two sample IDs (ONT037 and ONT37B) were combined; both contain reads from both basecaller models.

- **ont98**: 245 qualifying specimens from 837 total (environmental specimens). Basecalled with Dorado SUP v5.0.0, sequenced August 2025.

For ont98, qualifying specimens had a primary cluster with ≥ 200 reads representing $\geq 70\%$ of total reads. For ont37, thresholds were relaxed to ≥ 100 reads and $\geq 50\%$ dominance due to lower overall sequencing depth (median c1 read count of 59 vs 200+ for ont98).

2.2 HP length extraction

For each specimen, the SPOA multiple sequence alignment (MSA) was parsed to extract called hp lengths at each hp run in the consensus. The extraction algorithm:

1. Identify maximal hp runs in the ungapped consensus sequence.
2. For each run, locate the flanking non-hp bases (“anchors”) in the MSA.
3. For each read, count consecutive instances of the hp base between the anchors in the gapped alignment. This gives the read’s called hp length for that run.

2.3 Ground truth determination

Rather than using the SPOA consensus hp length as ground truth (which can be biased by SPOA’s heaviest-bundle algorithm), we used the **empirical mode** of called hp lengths across all reads at each position. At depths of 200+ reads on R10.4.1 SUP, the mode is the correct length for hp $\leq \sim 9$ [6].

2.4 Error classification

At each hp run, reads were classified as:

- **Correct**: called length equals the mode
- **HP deletion** ($-k$): called length is k bases shorter than the mode
- **HP insertion** ($+k$): called length is k bases longer than the mode
- **Interface substitution**: a base within the hp region is replaced by an adjacent flanking base (see Section 4)
- **Other substitution**: a non-flanking base appears within the hp region
- **Complex**: multiple error types within the hp region

Non-hp positions (singletons not adjacent to any run) were analyzed separately as a calibration baseline.

2.5 Scope of measurement

The error rates reported here are not raw nanopore sequencing error rates. They are **effective error rates at the post-clustering stage** of the speconsense pipeline: reads have been demultiplexed, clustered by MCL, and aligned with SPOA before analysis. Speconsense’s `--min-identity` parameter controls a quality filter during clustering that excludes reads below a pairwise identity threshold to the cluster consensus. Reads that fail this filter are never included in the MSA and therefore do not contribute to the error rates we measure.

This is the correct quantity for parameterizing the CER framework, which operates on the same post-clustering MSA. The effective error rate is what determines whether an observed variant in a speconsense cluster is plausibly explained by sequencing error. However, it means that our error rates are conditioned on speconsense’s clustering parameters and are not directly comparable to raw read-level error rates reported in the literature.

To assess the sensitivity of our results to this filtering, we repeated the analysis on ont37 and ont98 with `--min-identity 0.85` (vs the default of 0.90), which admits more divergent reads into clusters. Results are reported in Section 7.

3 Results (ont98)

3.1 Dataset summary

Metric	Value
Specimens analyzed	245
Total HP runs	129,673
Total read-level HP observations	83,429,870
Non-HP position observations	60,059,474

3.2 Non-HP error rates (calibration)

Error Type	Rate
Substitution	0.59%
Deletion	0.40%
Insertion	0.68%
Total (sub + del)	0.99%

These rates are consistent with R10.4.1 SUP read accuracy of $\sim 99\%$, confirming the calibration baseline.

3.3 HP error rates by base and length

Table 1 shows the fraction of reads calling the correct hp length (mode), and the fraction with ± 1 or larger deviations. Only contexts with ≥ 100 observations are shown.

Extended results for hp lengths 6–14 (all bases) are provided in the supplementary tables. Key trends at longer lengths: accuracy drops to 67–91% at length 7, 63–86% at length 8, and below 62% at length 10.

Table 1: HP error rates by base and length (ont98, raw).

Base	Length	N runs	N obs	Correct	Del (−1)	Ins (+1)	±2+
A	1	20,114	12,930,648	99.32%	0.55%	0.12%	0.01%
	2	8,479	5,487,156	98.25%	1.45%	0.21%	0.10%
	3	1,644	1,067,803	97.26%	2.13%	0.29%	0.32%
	4	595	378,225	96.15%	2.44%	0.41%	1.00%
	5	93	56,312	92.92%	4.57%	1.03%	1.48%
T	1	24,719	15,845,528	99.30%	0.54%	0.15%	0.01%
	2	7,337	4,770,094	98.05%	1.60%	0.24%	0.10%
	3	2,927	1,852,010	96.74%	2.53%	0.37%	0.37%
	4	657	430,793	95.60%	3.03%	0.60%	0.77%
	5	248	163,547	93.70%	3.89%	0.99%	1.42%
C	1	24,848	15,986,540	99.21%	0.69%	0.09%	0.00%
	2	4,770	3,053,272	98.44%	1.16%	0.26%	0.14%
	3	999	647,920	96.92%	2.15%	0.53%	0.40%
	4	309	189,751	95.47%	3.00%	0.85%	0.68%
	5	55	36,493	90.54%	5.60%	1.87%	1.99%
G	1	23,491	15,095,593	99.31%	0.60%	0.08%	0.01%
	2	6,858	4,476,352	98.38%	1.25%	0.21%	0.16%
	3	923	576,762	96.74%	2.26%	0.50%	0.50%
	4	245	147,437	93.99%	4.25%	0.81%	0.96%
	5	132	86,883	91.01%	5.58%	1.73%	1.68%

3.4 Implied per-position error rate

The run-level fraction correct compounds over the length of the hp run. To compare error rates across different hp lengths and against the non-hp baseline, we compute the implied per-position error rate p such that $(1 - p)^L$ equals the observed fraction correct. This decomposition is a convenience for cross-length comparison, not a mechanistic model: hp length errors are signal-level events affecting the entire run at once, not independent per-position errors. The per-position rate should be understood as a compact parameterization of the run-level error probability, useful for normalizing out the trivial effect of run length on aggregate accuracy.

Table 2: Pooled per-position error rates by HP length (ont98, raw).

Length	Run Error	Per-pos Rate	vs Non-HP (0.99%)
1	0.72%	0.72%	0.72×
2	1.74%	0.87%	0.88×
3	3.10%	1.04%	1.05×
4	4.45%	1.13%	1.14×
5	7.45%	1.54%	1.55×
6	11.03%	1.93%	1.95×
7	19.22%	3.00%	3.03×
8	28.92%	4.18%	4.22×
9	33.01%	4.35%	4.40×
10	38.13%	4.69%	4.73×

Key finding: At hp lengths 1–4, the implied per-position error rate is comparable to or below the non-hp baseline (0.7–1.1% vs 0.99%). This means that the run-level error probability at these lengths is within the range that would be expected if the per-position error rate were similar to the non-hp baseline—i.e., short hp runs are not inherently noisier than non-hp positions, once the effect of run length is accounted for. At lengths 5+, the implied per-position rate rises above the baseline, reaching 2–5× at lengths 7–9, reflecting the increasing difficulty of signal-level hp length estimation. (This per-position decomposition is a comparison metric, not a claim about the error mechanism; the full run-level error distributions are in Table 1 and in the supplementary `hp_distributions.tsv` tables.)

3.5 Qualitative observations

1. **Short hp runs (length 1–4) have very low error rates.** At length 1 (singletons), the hp “error” rate of 0.55–0.69% is comparable to the non-hp baseline of 0.99%. At length 4, accuracy is still 94–96%. These positions are strong candidates for CER-based variant recovery.
2. **Accuracy degrades monotonically with length, with a steep drop at length 7–8.** At length 7, accuracy ranges from 67% (G) to 91% (T). By length 10, accuracy is 54–62%—below the threshold where the mode is reliably the correct length.
3. **G/C homopolymers are harder than A/T.** G is consistently worst across all lengths. At length 5: A 92.9%, T 93.7%, C 90.5%, G 91.0%. The gap widens at longer lengths.
4. **Deletions dominate over insertions at all lengths.** The deletion : insertion ratio is 3–5 : 1 for moderate hp lengths (4–7), consistent with the known nanopore bias toward underestimating hp length.
5. **Our error rates are substantially better than Sereika et al. (2022).** At length 8, Sereika reported ~45–55% correct for R10.4 A/T reads; we observe 68–86%. This improvement likely reflects the newer basecaller (Dorado SUP v5.0.0 vs Guppy SUP) and possibly chemistry refinements (R10.4.1 vs R10.4), though the comparison is not exact: our effective error rates are conditioned on speconsense’s post-clustering MSA (Section 2.5), whereas Sereika measured raw read-level accuracy against a known reference.

3.6 SPOA consensus bias

Only 9 out of 129,673 hp runs (0.007%) showed a discrepancy between the SPOA consensus hp length and the empirical mode. The heaviest-bundle bias is negligible in this dataset, likely because the read depths are high enough that the mode strongly dominates.

3.7 Candidate biological hp variants

The bimodal distribution detector flagged 106 positions (0.08% of all hp runs) where a second hp length was present at rates significantly above the error model prediction. Many of these show approximately 50:50 splits consistent with heterozygous positions—variants that speconsense currently discards.

4 Interface Homopolymers

4.1 Definition

An “interface homopolymer” substitution occurs at the boundary between a homopolymer run and an adjacent base, where the substitution produces a sequence that can be reinterpreted as a hp length change on both adjacent runs. For example:

Consensus: ...TT|CC|TT...
Variant: ...TT|CT|TT... (C->T at position 2 of the CC run)

This can equivalently be read as: the T-run extended from 2 to 3, and the C-run shortened from 2 to 1. The adjusted-identity library treats this as hp-equivalent (0 edits), so speconsense does not detect it as a variant.

4.2 Prevalence

We classified every read observation at hp runs (length ≥ 2) by error type:

Error Type	Read Observations	Rate
Correct	22,585,613	95.77%
Pure HP deletion	249,385	1.06%
Pure HP insertion	~100,000	0.42%
Interface substitution	228,035	0.97%
Other substitution	122,236	0.52%
Complex	47,000	0.20%

Interface hp substitutions occur at **nearly the same rate as pure hp length errors** (0.97% vs 1.06%).

4.3 Expanded analysis: interface variant rate vs non-HP baseline

To determine whether interface hp substitutions are primarily signal-level artifacts or biological variants, we conducted an expanded analysis across 663 specimens (all specimens with ≥ 100 total reads, removing the original selection bias toward low-variation specimens).

Methodology: For each specimen, reads from all same-organism clusters ($\geq 90\%$ pairwise identity to the primary cluster consensus) were merged and re-aligned with SPOA to produce a combined MSA. This ensures that biological non-HP substitutions—which speconsense phases into separate clusters—are visible alongside interface HP substitutions, enabling a fair comparison. Each consensus position was classified as non-HP, HP interface (first or last base of a run of length ≥ 2 adjacent to a different base), or HP interior.

The key metric is the **fraction of positions showing a well-supported variant** (minor allele frequency $\geq$ threshold), compared across position classes. If interface positions produce artifactual variants due to signal-level boundary confusion, they should show an excess of well-supported variants relative to non-HP positions.

The rate ratio of interface to non-HP variant positions at MAF $\geq 20\%$ is **0.880 [95% bootstrap CI: 0.785–0.988]**, with the confidence interval entirely below 1.0. Interface positions produce well-supported variants at a slightly *lower* rate than non-HP positions.

Table 3: Variant position rates by position class (ont98, 663 specimens).

MAF Threshold	Non-HP	HP Interface	HP Interior	Ratio
$\geq 10\%$	0.319%	0.276%	0.328%	0.87
$\geq 20\%$	0.265%	0.231%	0.259%	0.87
$\geq 30\%$	0.219%	0.184%	0.199%	0.84
$\geq 40\%$	0.158%	0.133%	0.132%	0.84

Of the 457 interface positions with $\text{MAF} \geq 20\%$, only 173 (38%) had an interface-compatible top allele (i.e., the variant base matched the adjacent run’s base). This is close to the 33% expected by chance when there are three possible alternative bases, indicating no enrichment of boundary-confusion substitutions among well-supported variants.

Conclusion: There is no evidence that HP interface positions generate artifactual well-supported variants at rates above the non-HP baseline.

5 Biological variant filtering

The raw error rates above include HP positions where the observed variation is biological (genuine length polymorphisms) rather than sequencing error. The filtered analyses in subsequent sections (and the re-estimation in Section 8) identify and exclude these positions using two detectors:

- **Statistical outliers:** HP positions where the error rate significantly exceeds the pooled average for that context (one-sided binomial test, Bonferroni-corrected $\alpha = 0.01$).
- **Bimodal positions:** HP positions where a second allele is present at $> 2\times$ the expected error rate and $> 10\%$ of reads.

For ont98, the detectors flag 1,656 outlier positions and 106 bimodal positions (1,657 total excluded, 1.3% of HP runs). The filtered per-position rates are referenced as “ont98 filtered” in the cross-basecaller comparison (Section 6) and as the input distribution for re-estimation (Section 8).

6 Older Basecaller Comparison (ont37)

6.1 Dataset

To assess the effect of basecaller version on HP error rates, we analyzed ont37, a sequencing run from February 2024 basecalled with Dorado SUP v3.5.x (vs v5.0.0 for ont98). The data was combined from two sample IDs (ONT037 and ONT37B), with 70% of reads basecalled using the standard SUP v3.5.1 model and 30% using the 260bps SUP v3.5.2 model. Both datasets use R10.4.1 E8.2 chemistry.

Due to lower sequencing depth (median c1 read count of 59), qualifying thresholds were relaxed to ≥ 100 primary reads and $\geq 50\%$ dominance, yielding 94 qualifying specimens from 651 with c1 MSAs. Total filtered HP observations: 10.2 million (vs 82.0 million for ont98).

6.2 Non-HP error rates

The ont37 non-HP total error rate of 3.97% is $2.4\times$ higher than ont98. All error classes are elevated: substitutions $2.4\times$, deletions $2.5\times$, and insertions $2.3\times$.

Table 4: Non-HP error rates across both datasets.

Dataset	Substitution	Deletion	Insertion	Total
ont37 (v3.5)	1.44%	0.99%	1.54%	3.97%
ont98 (v5.0)	0.59%	0.40%	0.68%	1.67%

6.3 HP error rates and biological variant filtering

After biological variant filtering (817 outlier positions, 664 bimodal positions, 1,092 total excluded = 2.2%), the filtered per-position HP error rates across both datasets are:

Table 5: Filtered per-position HP error rates: cross-basecaller comparison.

HP Len	ont37 (v3.5)	ont98 (v5.0)	ont37/ont98
1	1.56%	0.67%	2.32
2	1.84%	0.83%	2.21
3	2.17%	0.97%	2.23
4	2.27%	0.99%	2.30
5	2.13%	1.13%	1.89
6	2.90%	1.57%	1.85
7	4.21%	2.23%	1.89

At L=1–5, ont37 error rates are consistently $\sim 2.2\times$ those of ont98, demonstrating substantial improvement from Dorado v3.5 to v5.0. Run-to-run variation within a basecaller version (independent of this v3.5→v5.0 effect) is expected but not characterized in this report.

6.4 Whole-run accuracy by HP length

Table 6: Whole-run accuracy (fraction correct) across datasets (filtered).

HP Length	ont37 (v3.5)	ont98 (v5.0)
1	98.4%	99.3%
2	96.4%	98.3%
3	93.6%	97.1%
4	91.2%	96.1%
5	89.8%	94.5%
6	83.8%	91.0%
7	74.0%	85.4%

Even with the older v3.5 basecaller, short HP accuracy remains high: 98.4% at L=1 and 91.2% at L=4. The CER framework can be applied to short HPs across basecaller versions, with the per-position error rate parameterized appropriately for each.

6.5 Base composition (ont37)

The base composition pattern in ont37 shows larger C/G vs A/T divergence at shorter lengths compared to the v5.0 dataset (max/min ratio of 1.54 at L=1 and 2.00 at L=5, vs 1.14 and 1.39 for

Table 7: Per-base variation in filtered HP error rates (ont37).

HP Length	A	C	G	T
1	1.26%	1.95%	1.71%	1.31%
2	1.79%	1.63%	1.82%	2.02%
3	1.61%	3.00%	2.35%	2.23%
4	2.32%	3.11%	3.17%	1.92%
5	2.11%	3.76%	2.37%	1.88%

ont98). This is consistent with the older basecaller having more difficulty with G/C homopolymers, a pattern that Dorado v5.0 substantially ameliorates.

6.6 Interface variant rate across basecaller versions

The expanded interface analysis (Section 4) repeated on ont37 (651 specimens evaluated, with cluster recombination):

Table 8: Interface vs non-HP variant rate ratio across both datasets.

Dataset	Specimens	Rate ratio	95% CI	Wilcoxon p
ont37	651	1.026	0.880–1.193	0.589
ont98	663	0.880	0.785–0.988	0.042

Both datasets are consistent: interface HP positions do not show excess variant rates relative to the non-HP baseline. The ont37 ratio of 1.026 has a confidence interval spanning 1.0, indicating no detectable effect even with the noisier v3.5 basecaller. The conclusion that interface positions require no special CER treatment holds across basecaller generations.

6.7 Base composition effects (ont98)

We examined whether error rates need to be segmented by base (A/T/C/G) at each HP length, using the ont98 filtered data:

Table 9: Per-base variation in filtered HP error rates (ont98).

HP Length	Max base	Min base	Max/Min
1	0.74% (C)	0.65% (A)	1.14
2	0.94% (T)	0.72% (C)	1.31
3	1.04% (T)	0.84% (A)	1.25
4	1.19% (G)	0.94% (A)	1.26
5	1.40% (G)	1.00% (A)	1.39
6	2.30% (G)	0.89% (A)	2.57
7	4.14% (C)	1.05% (T)	3.94

At L=1–5, the worst-case base is at most 1.4× the best—a modest penalty for using a single rate. Importantly, the “worst” base is not consistent (T at L=2–3, G at L=4–5), so there is no

stable A/T vs C/G split in this range. At $L \geq 6$, C/G diverges sharply from A/T (the well-established nanopore pattern), but sample sizes are small and these lengths are approaching the limit of reliable variant calling regardless.

7 Sensitivity to clustering identity threshold

Speconsense’s `--min-identity` parameter controls which reads are included in a cluster’s MSA. At the default value of 0.90, reads with $< 90\%$ pairwise identity to the cluster consensus are excluded. To test whether this quality filter materially affects our error rate estimates, we repeated the ont37 and ont98 analyses with `--min-identity 0.85`.

Table 10: Effect of `--min-identity` on filtered per-position HP error rates and non-HP error rates.

HP Len	ont37 (v3.5)		ont98 (v5.0)		085/default	
	default	0.85	default	0.85	ont37	ont98
1	1.56%	1.75%	0.67%	0.75%	1.12	1.11
2	1.84%	2.05%	0.83%	0.92%	1.11	1.10
3	2.17%	2.37%	0.97%	1.05%	1.10	1.08
4	2.27%	2.52%	0.99%	1.06%	1.11	1.08
5	2.13%	2.35%	1.13%	1.21%	1.10	1.07
Non-HP total error rate						
	3.97%	4.67%	1.67%	1.88%	1.18	1.13

Relaxing the identity threshold inflates error rates by $\sim 10\text{--}12\%$ at $L=1\text{--}5$, with a slightly larger effect on ont37 (where the older basecaller produces more reads near the identity boundary). The inflation is uniform across HP lengths, indicating that the additional reads are generally noisier rather than selectively worse at homopolymers.

Crucially, the basecaller scaling factor is unaffected: the ont37/ont98 ratio at `--min-identity 0.85` is 2.33 at $L=1$ and 2.37 at $L=4$, compared to 2.32 and 2.30 at the default threshold. The identity filter and basecaller version act as independent, multiplicative effects on the effective error rate.

For CER parameterization, the error rate should be estimated from data processed with the same `--min-identity` setting that will be used in production. The results in this report use the default value of 0.90 unless otherwise noted.

8 Re-estimation under updated pipeline

The original analysis (Sections 4–7) parameterized the shipped `dorado-v5.0.yaml` q_{ctx} table used by the context-aware CER framework in production [8]. The speconsense pipeline has since gained several mechanisms—context-aware variant phasing at $L \leq 5$ HP positions, cross-cluster read re-assignment, discard recovery, identity-group structure, and `err_factor`-based cluster homogeneity gating—each of which could shift measured error rates. We re-estimated rates under the current pipeline (speconsense 0.7.7, 2026-04-21) on the same ont98 input data to validate or update the shipped values.

8.1 Pipeline changes and expected directional effects

Table 11 summarizes the relevant changes and their expected effect on measured error rates. The changes fall into three classes: cluster-cleaning (which should *lower* apparent rates by producing cleaner single-template primaries), noise-admitting (which should *raise* rates by bringing in reads that used to be rejected), and selection-redefining (which change which specimens qualify for analysis without directly changing rates).

Table 11: Pipeline changes and their expected effect on measured HP/non-HP error rates. “↓” = expected to lower rates, “↑” = raise, “○” = neutral.

Change	Direction	Mechanism
Context-aware HP phasing ($L \leq 5$)	↓	Splits HP-only paralogs into siblings
Cross-cluster read reassignment	↓	Reads move to best-matching cluster at variants
Tighter <code>--min-variant-count</code> (5→3)	↓	More aggressive non-HP phasing
Linear SPOA scoring (match +1, else -1)	↓	Unbiased alignment improves variant phasing
<code>err_factor</code> cluster gating	↓	Selects self-consistent clusters
Discard recovery	↑	Re-admits initially-discarded reads
Identity group / complete linkage	○	Affects CER decisions, not q_{ctx}
Ambiguity calling (IUPAC)	○	Few positions affected in practice

8.2 Methods

ont98 (955 specimens) was re-processed with the current pipeline defaults, preserving only the sampling configuration that matches the original analysis: `--presample 0 --max-sample-size 10000 --collect-discards --error-model dorado-v5.0`. All other parameters took current defaults (`--min-identity 0.9, --group-identity 0.85, --hp-normalization-length 6, --min-variant-count 3, --significance-level 1e-5`). The pipeline also uses a linear SPOA scoring scheme (match +1, mismatch -1, gap open -1, gap extend -1) in place of SPOA’s defaults, which bias toward substitutions over gaps. No failures.

Selection criterion evolution. The original analysis selected specimens whose primary cluster had ≥ 200 reads and represented $\geq 70\%$ of total reads (“primary_frac” criterion). Under the more aggressive phasing of the current pipeline, this criterion becomes unworkable: on ont98 it drops from 245 to 157 qualifying specimens because cluster counts rise (median 10 clusters per specimen, with many small siblings pulling the primary fraction below 0.70). We therefore switched to an `err_factor`-based criterion: the primary anchor cluster must have `err_factor` < 1.0 and `RiC` ≥ 200 . This yields 445 qualifying specimens on ont98. The criterion is peer-independent (does not depend on sibling structure), directly measures single-template plausibility, and is stable under the current pipeline.

In practice the `err_factor` gate is non-binding at `RiC` ≥ 200 : all 445 qualifying ont98 specimens already satisfy `err_factor` < 1.0 (maximum observed value 0.929). `RiC` is therefore the effective selection criterion; `err_factor` serves here as a backup guarantee rather than an active filter. This distinction matters for the circularity argument: because q_{ctx} enters the selection only through `err_factor`, and `err_factor` is not constraining our pool, the rate estimates are not circularly dependent on the q_{ctx} value used to compute them. Section 8.6 confirms this analytically under the proposed table revision.

Approach 2: all-cluster non-HP analysis. To characterize the operational non-HP rate CER evaluates against in production (which includes candidate clusters of all sizes), we additionally ran a weighted analysis across all clusters with $\text{RiC} \geq 5$, using each cluster’s own consensus as its local scaffold. Results are stratified by RiC bin.

8.3 Results: rate reduction under the updated pipeline

Table 12 compares the original filtered rates ($\text{primary_frac} \geq 0.70$ criterion on old-pipeline clusters) to the re-estimated rates ($\text{err_factor} < 1.0$ criterion on new-pipeline clusters). Across all contexts, measured rates are consistently lower under the current pipeline, by a factor of approximately 0.70–0.80.

Table 12: Filtered per-position error rates on ont98: original analysis vs. re-estimation under the current pipeline. “v2_ef” denotes the current- pipeline rate under the $\text{err_factor} < 1.0$ selection.

Context	original	v2_ef
non-HP sub	0.0059	0.0039
non-HP del	0.0040	0.0028
non-HP ins	0.0068	0.0046
hp-l1	0.0067	0.0047
hp-l2	0.0083	0.0062
hp-l3	0.0097	0.0076
hp-l4	0.0099	0.0081
hp-l5	0.0113	0.0087

Framing observation. The current pipeline admits *more* reads into clusters than the original (via discard recovery and cross- cluster read reassignment). The a priori expectation under such additions is that per-cluster error rates should *rise*: more reads means more chances to admit noisier reads into any given cluster. The observed direction is the opposite—rates fall by 0.70–0.80 $\times$. The most parsimonious explanation is that the new cluster-composition mechanisms (reassignment and recovery placing reads into their best-matching clusters based on variant positions) produce cleaner per-cluster read populations than the original simpler MCL-only assignment, and that this cleaning effect dominates any noise admitted by expanding the read pool.

8.4 All-cluster non-HP rates by RiC bin

Table 13 reports the all-cluster pooled non-HP rates stratified by RiC . On ont98, rates are essentially flat across the RiC 20–1000+ range (substitution rate varies from 0.0036 to 0.0039). The RiC 5–20 bin is slightly elevated ($\sim 20\%$). The pooled non-HP rate (0.0039) is within 3% of the primary-anchor-only rate (0.0038), indicating that admitting small clusters does not materially shift the aggregate estimate.

8.5 Mechanism: isolating HP phasing contribution (Test A)

A natural concern is that the more aggressive HP phasing at $L \leq 5$ in the current pipeline could produce *artifactual* splits—cutting noise from signal and leaving a cleaner-than-reality primary cluster, thereby biasing apparent rates downward. To test this, we re-ran ont98 with

Table 13: All-cluster non-HP error rates by RiC bin (ont98). Each row pools reads across all MSAs (RiC ≥ 5) in the specimen, using per-cluster consensus as the ground-truth scaffold.

Bin (RiC)	Clusters	Sub	Sub+Del
5–20	5,472	0.0046	0.0078
20–50	632	0.0036	0.0058
50–100	324	0.0039	0.0063
100–200	289	0.0037	0.0062
200–500	408	0.0039	0.0066
500–1000	203	0.0038	0.0066
1000– ∞	29	0.0037	0.0065
POOLED	7,357	0.0039	0.0066

`--hp-normalization-length 0`, which reverts to the old behavior of blanket-normalizing all HP variants (no HP-driven splits).

Table 14 compares HP rates between the current-pipeline run (`v2_ef`) and the blanket-normalization variant (Test A). Rates shift by at most 0.0003 at $L \leq 5$. If HP phasing were producing substantial artifactual splits, disabling it would recover a large fraction of the gap to the original-pipeline rates; the observed gap recovery is only 2–11%.

Table 14: Effect of disabling HP phasing (`--hp-normalization-length 0`) on measured HP rates. Gap recovery is the fraction of the `v2_ef`→original gap closed by Test A.

L	v2_ef	Test A raw	Test A filt	filt delta	gap recovery
1	0.0047	0.0049	0.0048	+0.0000	2%
2	0.0062	0.0064	0.0062	+0.0000	2%
3	0.0076	0.0080	0.0076	+0.0001	3%
4	0.0081	0.0087	0.0082	+0.0001	6%
5	0.0087	0.0096	0.0090	+0.0003	11%

Two side-channel measurements confirm that HP phasing *is* doing real work, but that its output is absorbed by downstream filters:

- Cluster counts shifted under Test A: 387/955 specimens produced fewer clusters (HP-only splits now merged), 521 unchanged, 47 produced more. HP phasing is not inert; it is doing real partitioning.
- Bimodal flag counts (Step 6b of the biological-variant filter) doubled under Test A (22→46). When HP variants are merged back into the primary, they surface as bimodal length distributions, which the filter excludes from the filtered rate. The outlier-detection filter also caught an additional 122 positions.

The *filtered* rate aggregation is therefore robust to HP phasing behavior: merging HP variants back in doubles the bimodal flag count, the filter catches them, and the resulting per-context rates are nearly unchanged. This establishes that the bimodal and outlier filters (Section 4, *Biological variant filtering*) are a structural safety net for rate measurement, not a cosmetic post-hoc adjustment.

Implication for the rate drop. HP phasing accounts for approximately 2–11% of the measured rate reduction between the original and current pipelines. The remainder (~90%) comes from other changes, principally:

- **Cross-cluster read reassignment** (likely dominant): reads that were initially placed in a suboptimal cluster are moved to the cluster whose consensus they actually match at variant positions. Apparent per-read disagreement within each cluster falls accordingly. This mechanism changes cluster *membership*, not structure, so it is not isolated by Test A.
- **Linear SPOA scoring** (match +1, mismatch -1, gap open -1, gap extend -1) in place of SPOA’s defaults. The defaults bias toward substitutions over gaps, which forces reads with real indels into substitution-heavy alignments against the consensus and blurs the signal that variant phasing uses to separate paralogs. The linear scheme treats substitutions and indels equivalently, so indel-distinguished haplotypes phase cleanly and the resulting primary clusters are more homogeneous.
- More aggressive non-HP phasing (`--min-variant-count 5→3`) separating non-HP paralogs into siblings.
- Selection criterion change (`primary_frac` to `err_factor`) admitting a different specimen population.

Distinguishing among these contributions would require further test runs with individual flags reverted. For the purpose of this re-estimation, the relevant conclusion is that the observed rates are *not* inflated by artifactual HP splitting.

8.6 Revised shipped q_{ctx} table

Table 15 gives the proposed update to `dorado-v5.0.yaml`. HP rates use the `v2_ef` primary-anchor values (approach-1 regime, where mode-based ground truth is most reliable). Non-HP rates use the all-cluster pooled values from approach 2, which match the operational distribution CER evaluates against.

Table 15: Proposed revised q_{ctx} values for `dorado-v5.0.yaml`. “Original” values are the current shipped table (from the original analysis); “Revised” are the re-estimated values from this section.

Context	Original	Revised	Basis
<code>non-hp-sub</code>	0.0059	0.0039	approach 2 pooled (ont98)
<code>non-hp-indel</code>	0.0108	0.0074	approach 2 pooled (ont98)
<code>hp-11</code>	0.0067	0.0047	<code>v2_ef</code> primary (ont98)
<code>hp-12</code>	0.0083	0.0062	”
<code>hp-13</code>	0.0097	0.0076	”
<code>hp-14</code>	0.0099	0.0081	”
<code>hp-15</code>	0.0113	0.0087	”

The revised values shift `cer_factor` outputs upward by approximately $1/0.72 \approx 1.39\times$ on existing data. No upstream re-runs are required to adopt the new table; per the metadata-2.0 schema design [8] §6, stored output can be recomputed from the retained N , M , K , and context tags.

We confirmed analytically that the revised table does not induce a selection shift at our analysis tier: `err_factor` values scale approximately as $1/q_{\text{ctx}}$, so the revision inflates primaries' `err_factor` by $\sim 1.39\times$. The maximum observed primary `err_factor` in the current re-estimation (0.929) rises to ≈ 1.29 under the revised table—still below the default `--max-err-factor 1.5` filter threshold. Selection is therefore stable; no bootstrap iteration is required.

8.7 ont37 retune (Dorado v3.5)

The same procedure was applied to ont37 to refresh `dorado-v3.5.yaml`. Re-clustering was performed under speconsense 0.8.1 (released after the v5.0 retune) with `--error-model dorado-v3.5` (chemistry-matched to avoid underestimating the noise floor).

Selection criterion. ont37's lower per-specimen depth (median primary RiC was 59 in the original analysis) interacts with the more aggressive context-aware phasing of 0.8.x: primaries shrink further, and $\text{RiC} \geq 200$ yields only 10 specimens. We relaxed the floor to $\text{RiC} \geq 50$, yielding 80 qualifying specimens (vs. 94 under the original $\text{RiC} \geq 100$, `primary_frac` ≥ 0.50 criterion). The `err_factor` < 1.0 gate is empirically *fully* non-binding at this threshold (maximum observed value 0.737), preserving the circularity defense for the q_{ctx} -driven selection. The revised table inflates this maximum to ≈ 1.32 under $1/q_{\text{ctx}}$ scaling—still below the `--max-err-factor 1.5` cap—so selection remains stable after the table refresh.

Results. Table 16 compares the original ont37 rates to the v2_ef re-estimation. Rate ratios are broadly similar to ont98's ($0.56\text{--}0.74\times$ on ont37 vs. $0.66\text{--}0.82\times$ on ont98), confirming that the pipeline mechanisms responsible for the v5.0 rate drop apply to v3.5 data as well. The v3.5/v5.0 ratio narrows somewhat at higher L under the new pipeline ($2.23\times$ at $L = 1$ down to $1.41\times$ at $L = 5$, vs. a more uniform $\sim 2.2\times$ under the original) but remains well above unity at every short HP length.

Table 16: Filtered per-position error rates on ont37: original analysis vs. re-estimation under speconsense 0.8.1 (80 qualifying specimens at $\text{RiC} \geq 50$, `err_factor` < 1.0). Non-HP rates from approach 2 (all-cluster pooled, $\text{RiC} \geq 20$, 184 clusters).

Context	Original	v2_ef	Ratio
non-hp-sub	0.0144	0.0081	0.56
non-hp-indel	0.0222	0.0158	0.71
hp-11	0.0156	0.0105	0.67
hp-12	0.0184	0.0137	0.74
hp-13	0.0217	0.0150	0.69
hp-14	0.0227	0.0142	0.63
hp-15	0.0213	0.0123	0.58

Caveats specific to ont37. The qualifying pool is smaller than ont98's (80 vs. 445 specimens) and has lower per-specimen depth (RiC range 51–396 vs. ont98's ≥ 200). HP run counts at $L = 4\text{--}5$ are correspondingly limited (526 and 154 runs respectively), producing slight non-monotonicity in the rate-vs-length curve at high L . The bimodal/outlier filter exclusion rate is low (0.5%, 217/43,276 runs), indicating the filter's biological-variant cleanup is operating conservatively at this specimen

count. These values should therefore be treated as the best current estimates rather than as tightly constrained measurements; the test in `tests/test_qctx.py:test_v3_5_rates_higher_than_v5_0_at_short_length` remains satisfied (v3.5 rates exceed v5.0 by $1.75\times-2.23\times$ at $L = 1-4$).

Convergence (fixed-point check). Because the shipped table participates in the clustering step via the CER significance gate (through `err_factor` and variant phasing decisions), we verified that ont37 has converged to a fixed point of the bootstrap. ont37 was re-clustered a second time under the revised `dorado-v3.5.yaml` and the analysis re-run. Selection was highly stable (80 specimens both passes, 77 shared), and measured rates drifted by at most 2.5% relative: `hp-11` -2.40% , `hp-12` -2.49% , `hp-13` -1.46% , `hp-14` $+0.41\%$, `hp-15` $+0.36\%$; `non-hp-sub` -0.61% , `non-hp-indel` -0.66% . The mild downward direction at $L = 1-3$ is mechanistic: under the lower `q_ctx` expectations, the CER gate is more permissive at HP positions, so a few additional variants phase out into siblings and primaries are infinitesimally cleaner. All drifts sit within the sampling noise of the 80-specimen pool, so the shipped values are retained without further iteration. Median primary `err_factor` inflated from 0.541 to 0.731 between passes (less than the $1/q_{ctx}$ -only prediction of $\sim 1.78\times$, because cluster cleanup under the new `yaml` partially offsets the denominator drop); maximum observed value 1.092 remains well below the `--max-err-factor 1.5` cap.

8.8 Limitations

- **ITS-specific.** The re-estimation inherits the amplicon-specific limitation of the original analysis. Other amplicons or organism contexts may produce different values.
- **iNat input distribution.** The measurements reflect specimen quality distributions typical of iNaturalist uploads (ont98). Environmental DNA or specimen sources with substantially different input-quality variance could produce different all-cluster rates.
- **Run-to-run variation is not characterized in this report.** The shipped table is derived from a single ont98 run; runs of different provenance (other Mycota Lab batches, other operators) may produce systematically different rates. The CER factor's linear sensitivity to `q_ctx` [8] §3 handles this by design.
- **Mechanism attribution is not fully resolved.** Test A eliminates HP phasing as the dominant contributor. Definitive identification of the dominant mechanism (cross-cluster reassignment vs. non-HP phasing refinements vs. selection criterion) would require additional test runs with individual changes reverted; these are future work.
- **Phase-2 per-run estimation** ([8] §4.2) would eventually replace the shipped table entirely, eliminating the basecaller/chemistry-update dependency. The revised values are an interim update pending that work.

9 Discussion

9.1 Recommended approach for speconsense

The combined results from two sequencing runs spanning two basecaller generations support a simple two-tier approach:

Short homopolymers (length 1–5): Implied per-position error rates are consistently below 2.3% (filtered) across both datasets, all bases, and both basecaller versions. For the current Dorado

v5.0 basecaller, rates are below 1.4%. These positions can be brought into the CER framework—no special segmentation by base is needed (max/min ratio across bases is only 1.1–1.4× at these lengths for v5.0, and 1.2–2.0× for v3.5). The CER implementation should evaluate hp variants at the run level (i.e., “what is the probability of observing hp length k when the true length is L ?”) using the empirical error distributions from this analysis, rather than treating each position in the run as an independent trial. The per-position rate serves as a compact parameterization of the run-level error probability for a given (base, length) context, but the significance test itself operates on the run as a unit. The companion report [8] develops this into a deployed framework: per-basecaller tables of $q_{\text{ctx}}(\text{base}, \text{length})$ derived from this analysis feed the binomial survival test, and significance is reported as a dimensionless `cer_factor` (the ratio of the critical joint error rate to the product of empirical q_{ctx} values) that remains comparable across contexts and re-scales linearly as tables are refreshed for new chemistry or basecaller generations. The initial production deployment uses shipped per-basecaller tables pooled across base and direction; per-run estimation to account for flow-cell variation ([8] §4.2) is left as future work.

Long homopolymers (length 6+): Error rates rise steeply and diverge significantly by base (C/G 2–4× worse than A/T at $L=6-7$). Sample sizes are small, and the error rates are high enough that variant calling becomes unreliable. The current blanket normalization approach (treating all HP length differences as insignificant) remains appropriate for these lengths. Enhanced positional error reporting in the quality report could flag positions where the observed HP length distribution suggests a potential variant, even if the CER framework cannot confidently evaluate it.

Interface positions: No special treatment required. Well-supported variants at HP interface positions occur at or below the rate of non-HP variants, confirmed across both datasets (rate ratio 1.03 in ont37, 0.88 in ont98).

9.2 Interface homopolymers

The finding that interface hp substitutions are nearly as prevalent as pure hp length errors (0.97% vs 1.06%) initially raised the question of whether interface substitutions might be signal-level artifacts (boundary confusion in the nanopore signal). The expanded analysis resolves this question decisively across both datasets:

1. Well-supported variants at HP interface positions occur at or below the rate of non-HP variants (ont37 ratio 1.03 [95% CI: 0.88–1.19]; ont98 ratio 0.88 [95% CI: 0.79–0.99]). There is no excess that would indicate a signal-level artifact, even with the noisier v3.5 basecaller.
2. Among interface variant positions, the fraction with an interface-compatible top allele is close to the 33% expected by chance. Boundary-confusion substitutions are not enriched.
3. No published study has specifically analyzed substitution error rates at HP boundaries in nanopore data. Apparent boundary effects reported in the literature (e.g., Liu-Wei et al. [3], Dunn et al. [2]) are alignment artifacts from minimap2 representing HP length errors as SNPs at run boundaries, not signal-level errors.

9.3 Comparison to literature

Both datasets show substantially better HP accuracy than published R10.4 / Guppy SUP values [6], with progressive improvement across basecaller versions. Note that this comparison is approximate: Sereika measured raw read-level accuracy against a known reference genome, whereas our rates are effective error rates conditioned on speconsense’s post-clustering MSA (Section 2.5). The quality

filtering inherent in clustering excludes the noisiest reads, so our effective rates are expected to be somewhat lower than raw read-level rates for the same chemistry and basecaller. The magnitude of the difference attributable to this methodological difference vs. genuine basecaller improvement cannot be separated without matched raw-read measurements on our data.

Table 17: Whole-run accuracy progression from Sereika (Guppy) through our datasets (Dorado v3.5 and v5.0). Our values are effective post-clustering rates (Section 2.5), not directly comparable to Sereika’s raw read-level measurements.

HP Len	Sereika R10.4	ont37 (v3.5)	ont98 (v5.0)
1	~99.2%	98.4%	99.3%
3	~94.5%	93.6%	97.1%
5	~86.0%	89.8%	94.5%
7	~60.0%	74.0%	85.4%

The ont37 data (Dorado v3.5) provides an intermediate point between the published Guppy results and our v5.0 data, demonstrating a clear trajectory of improvement. Even at the v3.5 level, short HP accuracy is sufficient for CER analysis. The cross-basecaller comparison reveals a substantial version-level effect ($\sim 2.2\times$ from v3.5 to v5.0). Run-to-run variation within a basecaller version is plausible (flow cell quality, library preparation, basecaller calibration drift) but is not characterized in this report; per-run error rate estimation in production is left as future work.

9.4 Limitations and open questions

- Pipeline-conditioned measurement:** Our error rates are effective rates at the post-clustering stage, conditioned on speconsense’s identity threshold and clustering parameters (Section 2.5). They are not directly comparable to raw read-level error rates in the literature. The $\sim 10\text{--}12\%$ sensitivity to the identity threshold (Section 7) is modest but means that CER parameters should be estimated from data processed with the same settings used in production.
- Run-specific error rates:** The $\sim 2.2\times$ scaling factor between basecaller versions (v3.5 to v5.0) suggests that HP error rates should be estimated per-basecaller rather than assumed universal. Within-basecaller run-to-run variation (flow cell quality, library preparation, basecaller calibration drift) is expected but is not characterized in this report.
- Ground truth:** We use the empirical mode as ground truth, which is reliable for $hp \leq \sim 9$ at our read depths but may be incorrect for longer runs. Cross-platform validation with PacBio HiFi data would address this.
- PCR artifacts:** PCR can introduce hp length errors that are physically present in the library. These are indistinguishable from sequencing errors in our analysis and are out of scope for the CER framework (which establishes physical library presence, not biological reality).
- Sample composition:** Our data is exclusively ITS amplicons from fungi. The hp length distribution and context effects may differ for other loci or organisms, though the physical error mechanism should be universal.
- Biological variant filtering:** The outlier/bimodal detection thresholds are heuristic. A more sophisticated approach (e.g., cross-specimen consistency checks) could improve filtering, though the current approach is sufficient for the cross-basecaller comparison.

Acknowledgments

All sequencing data analyzed in this report were produced by Mycota Lab (Hoosier Mushroom Society) under the protocol of Russell [4]. The author thanks Stephen D. Russell for sharing the run-level outputs that made this analysis possible.

Substantial portions of the text were drafted with the assistance of Claude (Anthropic, Opus 4.6). The core ideas, analytical framing, and mathematical approach were provided by the author; the AI contributed to exposition, L^AT_EX formatting, and iterative drafting under human direction. The author takes full responsibility for the content.

References

- [1] C. Delahaye and J. Nicolas. Sequencing DNA with nanopores: Troubles and biases. *PLoS ONE*, 16(10):e0257521, 2021.
- [2] T. Dunn, E. Berry, C. M. Bergman. nPoRe: *n*-polymer read correction for improved nanopore homopolymer accuracy. *BMC Bioinformatics*, 24:292, 2023.
- [3] W. Liu-Wei, W. Trzaskowski, et al. Benchmarking Oxford Nanopore long-read RNA sequencing for transcript quantification. *BMC Genomics*, 25:219, 2024.
- [4] S. D. Russell. ONT DNA barcoding fungal amplicons w/ MinION & Flongle V.3. *protocols.io*, March 2023. <https://dx.doi.org/10.17504/protocols.io.36wgq7qykvk5/v3>.
- [5] Y. Ni, G. Liu, H. Peng, et al. Benchmarking of Nanopore R10.4 and R9.4.1 flow cells in single-cell whole-genome amplification and whole-genome shotgun sequencing. *Computational and Structural Biotechnology Journal*, 21:2352–2363, 2023.
- [6] M. Sereika, R. H. Kirkegaard, S. M. Karst, et al. Oxford Nanopore R10.4 long-read sequencing enables the generation of near-finished bacterial genomes from pure cultures and metagenomes without short-read or reference polishing. *Nature Methods*, 19:482–488, 2022.
- [7] J. Walker. Critical error rate analysis for MSA-based variant phasing in nanopore amplicon sequencing. Technical report, 2026. https://github.com/joshuaowalker/speconsense/blob/main/docs/variant_significance/variant_significance.pdf
- [8] J. Walker. Critical error rate in practice: context-aware variant significance for nanopore amplicon pipelines. Technical report, 2026. https://github.com/joshuaowalker/speconsense/blob/main/docs/cer_in_practice/cer_in_practice.pdf
- [9] R. R. Wick. Benchmarking Oxford Nanopore bacterial genome assembly. *Microbial Genomics*, 2025.